```
else:
    fig, ax = plt.subplots(
        2,
        1,
        figsize=(8, 6),
        sharex=True,
        gridspec_kw={"height_ratios": [2, 1]}
    )

    ...  ❶

    if next_x is not None:  ❷
        ax[0].axvline(next_x, linestyle="dotted", c="k")  ❷

    ax[1].plot(xs, acquisition_score, c="g")  ❸
    ax[1].fill_between(  ❸
      xs.flatten(),  ❸
      acquisition_score,  ❸
      0,  ❸
      color="g",  ❸
      alpha=0.5  ❸
    )  ❸

    if next_x is not None:  ❷
        ax[1].axvline(next_x, linestyle="dotted", c="k")  ❷

    ax[1].set_ylabel("acquisition score")

    plt.show()
```

❶ GP predictions; same as before.

❷ The point maximizing the acquisition score, visualized using a dotted vertical line.

❸ The acquisition score, plotted in a separate subplot.

This function, when `policy` and `next_x` are passed in, will create a lower panel shows the acquisition score according to the BayesOpt policy. Finally, we need to implement Step 3 of the BayesOpt loop in Figure 4.2, which (1) finds the point that has the highest acquisition score and (2) adds it to the training data and updates the GP.

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